



3

Number and Algebra

Coordinate geometry

Straight lines are an important part of our environment. We play sport on courts with parallel and perpendicular lines, and skyscrapers would not be standing without straight lines. We can also use straight lines to model different types of data and predict future outcomes.

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Chapter outline

	Proficiency strands			
3-01 Length, midpoint and gradient of an interval	U	F	R	C
3-02 Parallel and perpendicular lines	U	F	R	C
3-03 Graphing linear equations	U	F	R	C
3-04 The gradient–intercept equation $y = mx + b$	U	F	R	C
3-05 The general form of a linear equation $ax + by + c = 0$	U	F	R	C
3-06 The point–gradient form of a linear equation*	U	F	R	C
3-07 Finding the equation of a line	U	F	R	C
3-08 Equations of parallel and perpendicular lines	U	F	R	C

*STAGE 5.3

Wordbank

general form Any linear equation expressed as $ax + by + c = 0$, where a , b and c are integers and a is positive

gradient The steepness of a line or interval, measured by the fraction $\frac{\text{rise}}{\text{run}}$

gradient–intercept form Any linear equation expressed as $y = mx + b$, where m is the gradient and b is the y -intercept

linear equation An equation whose graph is a straight line

parallel lines Lines that point in the same direction and have the same gradient

perpendicular lines Lines that cross at right angles (90°) and have gradients whose product is -1

x -intercept The x -value at which a graph cuts the x -axis

y -intercept The y -value at which a graph cuts the y -axis

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Coordinate geometry

In this chapter you will:

- find the distance between two points located on the Cartesian plane using a range of strategies, including graphing software
- find the midpoint and gradient of a line segment (interval) on the Cartesian plane using a range of strategies, including graphing software
- sketch linear graphs using the coordinates of two points and solve linear equations
- solve problems involving parallel and perpendicular lines 
- (STAGE 5.3) use coordinate geometry formulas to calculate the length, midpoint and gradient of an interval
- (STAGE 5.3) find the angle of inclination of a line using the formula $m = \tan \theta$
- (STAGE 5.3) graph a line by finding its x - and y -intercepts
- test whether a point lies on a line
- use the gradient–intercept equation of a straight line $y = mx + b$
- find the equation of a line from its graph
- recognise the general form of the equation of a straight line and convert it to the gradient–intercept equation
- (STAGE 5.3) find the equation of a line given its gradient and a point on the line, or given two points, by using the point–gradient formula
- find the equation of a line that is parallel or perpendicular to a given line
- (STAGE 5.3) use coordinate geometry methods to prove geometrical properties

SkillCheck

Worksheet

StartUp assignment 2

MAT10NAWK10008

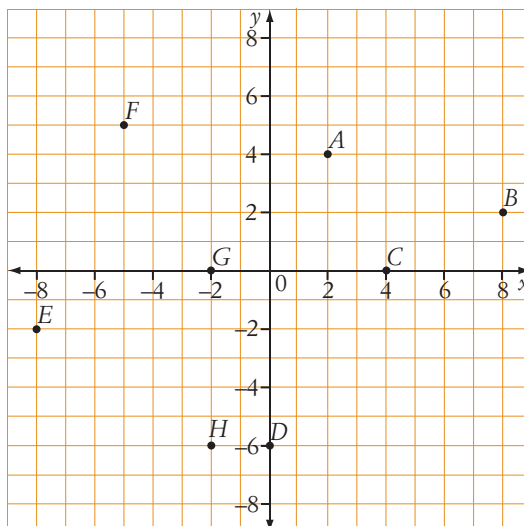
Skillsheet

Pythagoras' theorem

MAT10MGSS10004

1 For this number plane, find:

- | | |
|---|--|
| a the midpoint of interval BC | b the midpoint of interval HE |
| c the length of interval GC | d the length of interval GH |
| e the lengths of AC and BC ,
correct to one decimal place | f the type of triangle $\triangle ABC$ is |
| g the gradient of GE | h the gradient of EH |



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2 For each linear equation, copy and complete the table of values and graph the equation.

a $y = x - 3$

x	0	1	2	3
y				

b $y = 3x + 2$

x	-2	-1	0	1
y				

c $y = 1 - 2x$

x	-1	0	1	2
y				

3 If $x_1 = 3$, $y_1 = 4$, $x_2 = -5$ and $y_2 = 6$, then evaluate each expression.

a $x_1 + x_2$

b $x_2 - x_1$

c $\frac{y_1 + y_2}{2}$

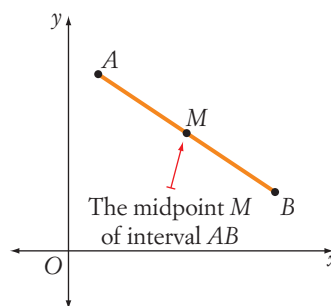
d $(y_2 - y_1)^2$

e $\frac{y_2 - y_1}{x_2 - x_1}$

f $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

3-01 Length, midpoint and gradient of an interval

The **length** of an interval AB (or the **distance** between A and B) can be calculated using Pythagoras' theorem if we know the coordinates of A and B .

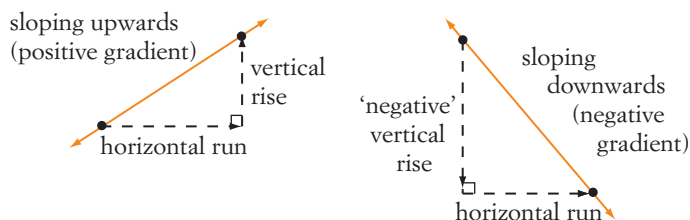


The **midpoint** of an interval AB is the point in the middle of AB or halfway between A and B .

- Its x -coordinate is the average of the x -coordinates of A and B .
- Its y -coordinate is the average of the y -coordinates of A and B .

The **gradient** of an interval measures its steepness. It is given by the formula:

$$m = \frac{\text{vertical rise}}{\text{horizontal run}} = \frac{\text{rise}}{\text{run}}$$



- A line **sloping upwards** has a **positive rise** and a **positive gradient**.
- A line **sloping downwards** has a **negative rise** and a **negative gradient**.
- The **run** is always **positive**.

Worksheet

Gradient, midpoint, distance

MAT10NAWK00014

Puzzle sheet

Intervals match-up

MAT10NAPS10009

Technology worksheet

Excel worksheet: Midpoint and distance between two points

MAT10NACT00008

Technology worksheet

Excel spreadsheet: Midpoint and distance

MAT10NACT00038

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 Coordinate geometry

Example 1

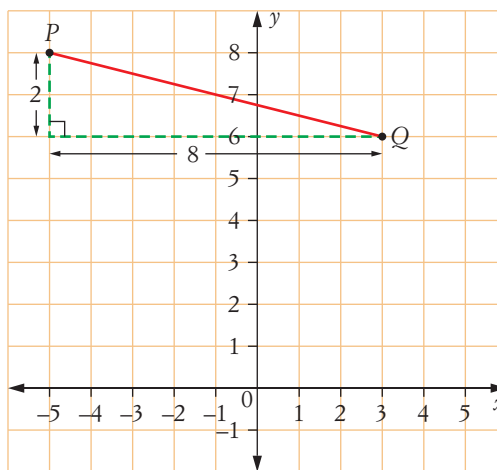
For the interval joining the pair of points $P(-5, 8)$ and $Q(3, 6)$, find:

- the length of the interval, correct to one decimal place
- the midpoint of the interval
- the gradient of the interval

Solution

- Draw a right-angled triangle on the number plane with PQ as the hypotenuse.

The height of the triangle is 2 units.
 The base of the triangle is 8 units.



$$PQ^2 = 2^2 + 8^2$$

$$= 68$$

$$PQ = \sqrt{68}$$

$$= 8.2462 \dots$$

$$\approx 8.2 \text{ units}$$

by Pythagoras' theorem

- For $P(-5, 8)$ and $Q(3, 6)$, the average of the x -coordinates is $\frac{-5+3}{2} = -1$.
 The average of the y -coordinates is $\frac{8+6}{2} = 7$.
 $\therefore$ The midpoint of PQ is $(-1, 7)$.

From the diagram above, a midpoint at $(-1, 7)$ looks reasonable.

- The rise is -2 units.
 The run is 8 units.

$$m = \frac{\text{rise}}{\text{run}}$$

$$= \frac{-2}{8}$$

$$= -\frac{1}{4}$$

Line slopes downwards.

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The distance, midpoint and gradient formulas

The methods for finding the length, midpoint and gradient of an interval can each be summarised by a formula.

The **distance formula** is used to calculate the distance (d) between any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$, in other words, the length of the interval PQ .

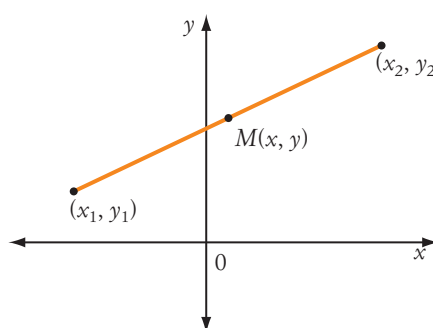
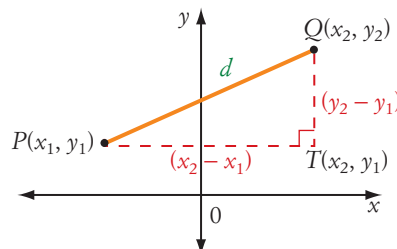
$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

by Pythagoras' theorem

The **midpoint formula** gives the coordinates of the point M , the midpoint of the interval joining $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$



The **gradient formula** gives the gradient of the interval or line joining P and Q .

$$\text{Gradient, } m = \frac{\text{rise}}{\text{run}} = \frac{\text{difference in } y}{\text{difference in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Summary

For an interval PQ with endpoints $P(x_1, y_1)$ and $Q(x_2, y_2)$, the formulas for distance (length), midpoint and gradient are:

Distance $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint $M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$

Example 2

For the interval joining $P(-5, 8)$ and $Q(3, 6)$ from Example 1b, use a formula to find:

- the length of the interval, correct to one decimal place
- the midpoint of the interval
- the gradient of the interval.

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Video tutorial

Coordinates geometry

MAT10NAVT00005

Video tutorial

Distance, midpoint and gradient formulas

MAT10NAVT10010

Puzzle sheet

Finding coordinates for given segment lengths

MAT10NAPS00048

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Coordinate geometry

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Solution

For $P(-5, 8)$ and $Q(3, 6)$:

$$\begin{array}{cc} \uparrow & \uparrow \\ (x_1, y_1) & (x_2, y_2) \end{array}$$

$$\begin{aligned} \text{a } d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(3 - (-5))^2 + (6 - 8)^2} \\ &= \sqrt{68} = 8.2462 \dots \approx 8.2 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{b } M(x, y) &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{-5 + 3}{2}, \frac{8 + 6}{2} \right) \\ &= (-1, 7) \end{aligned}$$

$$\begin{aligned} \text{c } m &= \frac{\text{difference in } y}{\text{difference in } x} \\ &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{6 - 8}{3 - (-5)} = \frac{-2}{8} = -\frac{1}{4} \end{aligned}$$

$$x_1 = -5, y_1 = 8, x_2 = 3, y_2 = 6$$

Apply the distance formula.

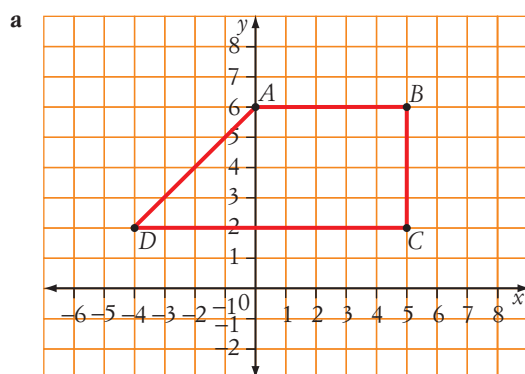
Apply the midpoint formula.

Apply the gradient formula.

Example 3

- Plot the points $A(0, 6)$, $B(5, 6)$, $C(5, 2)$ and $D(-4, 2)$ on a number plane and join them to make the quadrilateral $ABCD$.
- What type of quadrilateral is $ABCD$?
- Find the exact length of AD .
- Hence find the perimeter of $ABCD$ correct to two decimal places.

Solution



Join the points in the correct order.

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b Since $AB \parallel CD$, the quadrilateral is a trapezium.

c $AD^2 = 4^2 + 4^2 = 32$

$AD = \sqrt{32}$ units

In exact surd form.

d By counting grid squares, $AB = 5$, $BC = 4$, $CD = 9$.

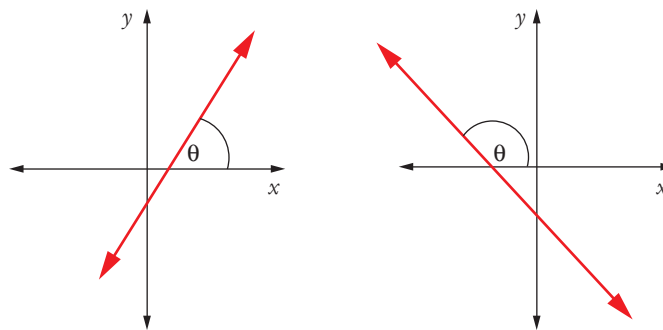
Perimeter of $ABCD = 5 + 4 + 9 + \sqrt{32} = 23.656 \dots \approx 23.66$ units

The angle of inclination of a line

The **angle of inclination**, θ , of a line is the angle it makes with the x -axis in the positive direction.

acute angle = positive gradient

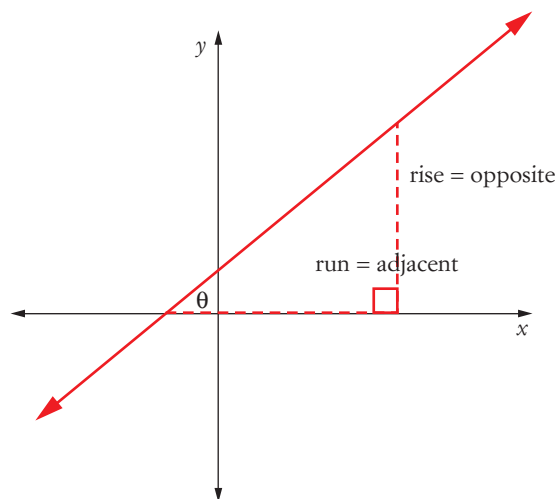
obtuse angle = negative gradient



Note from the above diagrams that θ is **acute** when the line has a **positive gradient**, and **obtuse** when the line has a **negative gradient**.

We can use trigonometry to calculate the angle of inclination of a line using its gradient, m .

The diagram below shows that $m = \frac{\text{rise}}{\text{run}}$, but in trigonometry, $\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\text{rise}}{\text{run}}$.
 $\therefore m = \tan \theta$.



Summary

The **angle of inclination**, θ , of a line is related to the gradient, m , of the line by the formula:

$$m = \tan \theta$$

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Example 4

Find, correct to the nearest degree, the angle of inclination of a line with gradient:

a $\frac{1}{3}$ b -4

Solution

a $m = \tan \theta$

$$\frac{1}{3} = \tan \theta$$

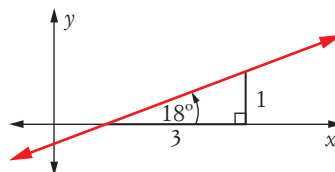
$$\tan \theta = \frac{1}{3}$$

$$\theta = 18.4349 \dots$$

$$\approx 18^\circ$$

The positive gradient means that it is an acute angle

On a calculator: **SHIFT** **tan** **1** **a/b/c** **3** **=**



b $m = \tan \theta$

$$-4 = \tan \theta$$

$$\tan \theta = -4$$

$$\theta = -75.9637 \dots$$

$$\approx -76^\circ$$

But this negative angle is the angle below the x -axis.

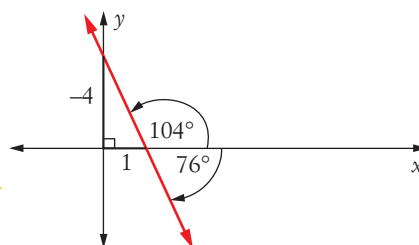
To find the angle of inclination,

$$\theta \approx 180^\circ - 76^\circ$$

$$\approx 104^\circ$$

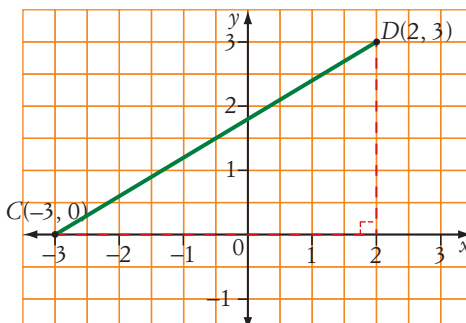
The negative gradient means that it is an obtuse angle

On a calculator: **SHIFT** **tan** **(-)** **4** **=**



Exercise 3-01 Length, midpoint and gradient of an interval

Questions 1, 2 and 3 refer to this diagram of interval CD .



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- 1 What is the length of interval CD ? Select the correct answer **A**, **B**, **C** or **D**.

A 2 units **B** 5.8 units **C** 3.2 units **D** 8 units

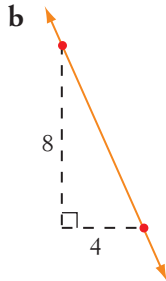
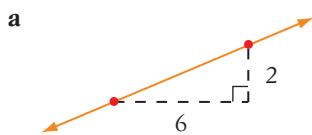
- 2 What is the midpoint of CD ? Select **A**, **B**, **C** or **D**.

A $(-1, 3)$ **B** $(-5, 3)$ **C** $(-0.5, 1.5)$ **D** $(-2.5, 1.5)$

- 3 What is the gradient of CD ? Select **A**, **B**, **C** or **D**.

A $\frac{3}{5}$ **B** -3 **C** $-\frac{5}{3}$ **D** 2

- 4 Calculate the gradient of each line.



- 5 For the interval joining each pair of points given, find:
 i the length of the interval correct to one decimal place

ii the midpoint of the interval

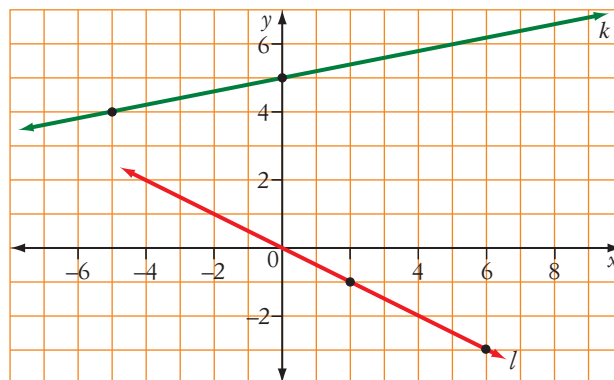
iii the gradient of the interval.

- a** $A(5, 3)$ and $B(7, 2)$ **b** $J(-1, 0)$ and $K(8, 6)$ **c** $M(0, -3)$ and $N(-5, 2)$
d $R(-3, -6)$ and $S(4, -9)$ **e** $A(-7, 2)$ and $B(-5, -8)$ **f** $U(3, -2)$ and $V(7, 2)$

- 6 Calculate, in exact (surd) form, the distance between each pair of points.

- a** $(-8, -1)$ and $(0, 4)$ **b** $(12, -6)$ and $(-1, -1)$ **c** $(7, -2)$ and $(-2, -3)$

- 7 Find the gradient of the lines labelled k and l .



- 8 Which expression gives the y -coordinate of the midpoint of the interval joining points $(3, 8)$ and $(-1, 5)$? Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{-1+5}{2}$ **B** $\frac{8+5}{2}$ **C** $\frac{8-5}{2}$ **D** $\frac{5-8}{2}$

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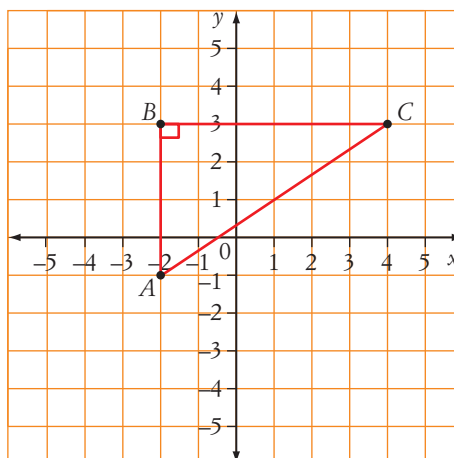
Stage 5.3

See Example 3

- 9 The vertices of triangle ABC are $A(-1, -1)$, $B(1, 3)$ and $C(3, 1)$.
 - a Draw $\triangle ABC$ on a number plane.
 - b Find the exact length of each side of the triangle.
 - c Are any sides of the triangle equal in length?
 - d What type of triangle is ABC ?
 - e Find the perimeter of $\triangle ABC$, correct to one decimal place.
- 10 The vertices of quadrilateral $KLMP$ are $K(1, 6)$, $L(7, 2)$, $M(3, -4)$ and $P(-3, 0)$.
 - a Draw the quadrilateral on a number plane.
 - b What type of quadrilateral is $KLMP$?
 - c Find the gradients of sides KL and PM .
 - d Find the gradients of sides KP and LM .
 - e What do you notice about the gradients of opposite sides of this quadrilateral? What does that mean about those sides?
 - f Find the exact length of each side of $KLMP$.
 - g Find the perimeter of $KLMP$, correct to one decimal place.
 - h Find the area of $KLMP$.

- 11 This diagram shows a right-angled triangle with vertices $A(-2, -1)$, $B(-2, 3)$ and $C(4, 3)$.

- a Copy the diagram and find the coordinates of P and Q , the midpoints of BA and BC respectively. Mark P and Q on your diagram.
- b Calculate, correct to one decimal place, the lengths of PQ and AC . What do you notice about your answers?
- c Find the gradients of PQ and AC . What do you notice about your answers?



See Example 4

- 12 Find, correct to the nearest degree, the angle of inclination of a line with gradient:

a 3	b $\frac{1}{2}$	c 1	d 2.5
e -2	f $\frac{3}{4}$	g $-\frac{1}{10}$	h $-\frac{2}{3}$
- 13 Find, correct to two decimal places, the gradient of a line with angle of inclination:

a 60°	b 158°	c 42°	d 94°
e 8°	f 135°	g 177°	h 0°

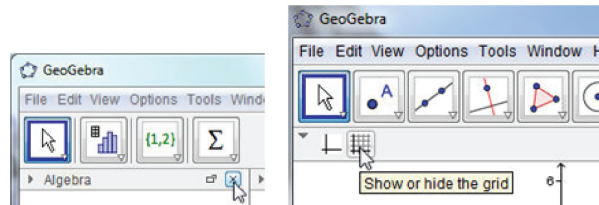
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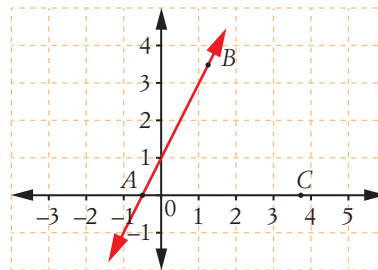
Technology The angle of inclination

In this activity we will use GeoGebra to calculate the angle of inclination of a line.

- 1 Close the **Algebra** View so that only the graphics window is showing and select the grid option at the top left-hand corner. Click on the input bar at the bottom of the screen and enter: $y = 2x + 1$



- 2 Click New Point. Click on the x -intercept of the line $y = 2x + 1$ (where it meets the x -axis). Also place New Points on the straight line (shown below as B) and the x -axis (shown below as C).



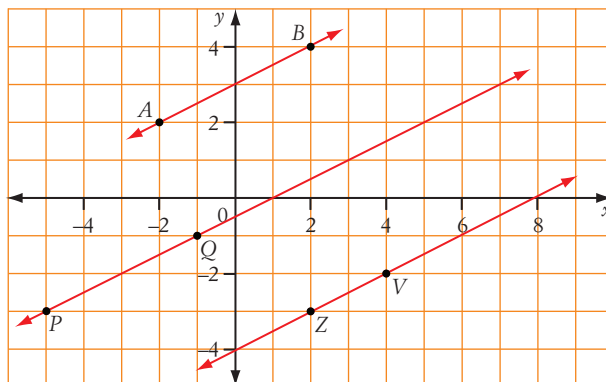
- 3 Click Angle and select in a clockwise direction the points C , A and B in order.
- 4 What is the angle of inclination of the line? Answer to the nearest degree.
- 5 Use GeoGebra to measure the angle of inclination of the line with equation:

a $y = -3x - 5$	b $y = x + 2$	c $y = x - 6$	d $y = -2x + 3$
e $y = -5x - 7$	f $y = -8x + 1$	g $y = 3x - 12$	h $y = \frac{2}{3}x + 4$

Investigation: Parallel and perpendicular lines

- 1 These three lines are parallel. Calculate the gradient of:

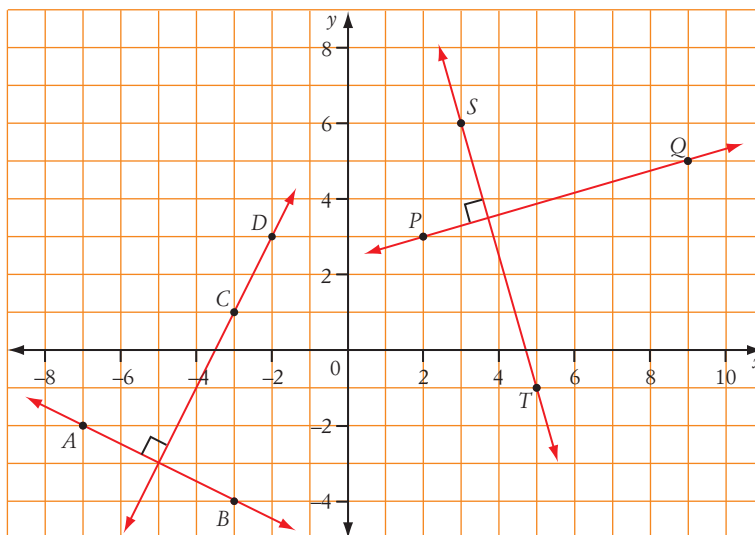
a AB b PQ c ZV



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- 2 What can you conclude about the gradients of parallel lines?
- 3 This diagram shows two pairs of perpendicular lines. $AB \perp CD$ and $PQ \perp ST$.



Calculate the gradient of:

- a AB b CD c PQ d ST

- 4 Is there a relationship between:
 - a the gradients of AB and CD ?
 - b the gradients of PQ and ST ?
- 5 Calculate the product of (multiply):
 - a the gradients of AB and CD
 - b the gradients of PQ and ST
- 6 What can you conclude about the gradients of perpendicular lines?

Puzzle sheet

Gradients of parallel and perpendicular lines

MAT10NAPS00012

Technology

GeoGebra:
Perpendicular lines

MAT10NATC00005

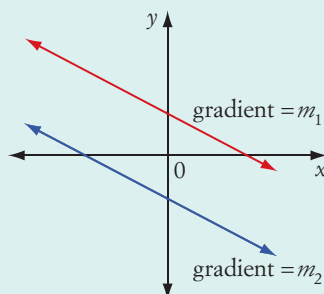
3-02 Parallel and perpendicular lines

Parallel lines

Summary

Parallel lines have the same gradient.

If two lines with gradients m_1 and m_2 are parallel, then $m_1 = m_2$



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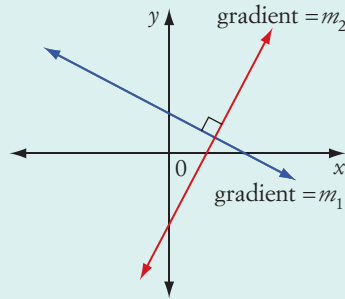
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Perpendicular lines

Summary

Perpendicular lines have gradients whose product is -1 .

If two lines with gradients m_1 and m_2 are perpendicular, then $m_1 \times m_2 = -1$ or $m_2 = -\frac{1}{m_1}$.



Note that m_2 is the **negative reciprocal** of m_1 .

Example 5

State whether each pair of gradients represent parallel lines, perpendicular lines or neither.

a $m_1 = \frac{1}{2}, m_2 = 2$ **b** $m_1 = 0.4, m_2 = \frac{2}{5}$ **c** $m_1 = 1\frac{3}{5}, m_2 = -\frac{5}{8}$

Solution

a $m_1 \neq m_2$ so the lines are not parallel.

$$\begin{aligned} m_1 \times m_2 &= \frac{1}{2} \times 2 \\ &= 1 \\ &\neq -1 \end{aligned}$$

so the lines are not perpendicular.

$\therefore$ The lines are neither parallel nor perpendicular.

c $m_1 = 1\frac{3}{5} = \frac{8}{5}$

$$\begin{aligned} m_1 \times m_2 &= \frac{8}{5} \times \left(-\frac{5}{8}\right) \\ &= -1 \end{aligned}$$

$\therefore$ The lines are perpendicular.

b $m_2 = \frac{2}{5} = 0.4$

$$m_1 = m_2$$

$\therefore$ The lines are parallel.

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Example 6

Find the gradient of a line that is perpendicular to a line with gradient:

- a 2 b -3 c $\frac{3}{4}$ d -0.6

Solution

a $m_1 = 2$

$$m_2 = \frac{-1}{m_1} \text{ for perpendicular lines}$$

$$= \frac{-1}{2}$$

$$= -\frac{1}{2}$$

The gradient is $-\frac{1}{2}$.

c $m_1 = \frac{3}{4}$

$$m_2 = \frac{-1}{m_1}$$

$$= \frac{-1}{(\frac{3}{4})}$$

$$= -\frac{4}{3}$$

The gradient is $-\frac{4}{3}$.

b $m_1 = -3$

$$m_2 = \frac{-1}{m_1} \quad \text{The negative reciprocal of } m_1.$$

$$= \frac{-1}{-3}$$

$$= \frac{1}{3}$$

The gradient is $\frac{1}{3}$.

d $m_1 = -0.6 = -\frac{3}{5}$

$$m_2 = \frac{-1}{(-\frac{3}{5})}$$

$$= \frac{5}{3}$$

The gradient is $\frac{5}{3}$.

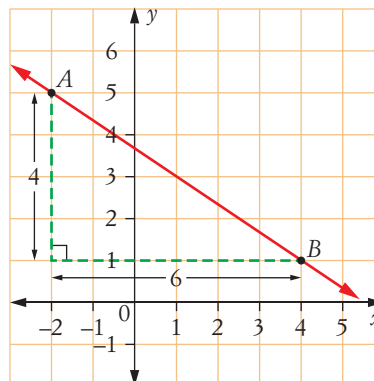
Example 7

A line passes through the points $A(-2, 5)$ and $B(4, 1)$. What is the gradient of a line:

- a parallel to AB ? b perpendicular to AB ?

Solution

Find the gradient of AB by calculating the rise and run.



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$$\text{Rise} = 1 - 5 = -4$$

$$\text{Run} = 4 - (-2) = 6$$

$$\text{Gradient } AB = \frac{-4}{6} = -\frac{2}{3}$$

Difference between y -coordinates.

Difference between x -coordinates.

$\frac{\text{rise}}{\text{run}}$

- a** Any line parallel to AB will have the same gradient as AB .

$$\therefore m = -\frac{2}{3}$$

- b** The gradient of a line perpendicular to AB will be given by:

$$m = \frac{-1}{(-\frac{2}{3})} = \frac{3}{2}$$

Exercise 3-02 Parallel and perpendicular lines

- 1** State whether each pair of gradients represent parallel lines, perpendicular lines or neither.

a $m_1 = \frac{1}{4}, m_2 = 4$ **b** $m_1 = 3, m_2 = -\frac{1}{3}$ **c** $m_1 = 0.5, m_2 = \frac{1}{2}$

d $m_1 = \frac{2}{7}, m_2 = \frac{7}{2}$ **e** $m_1 = \frac{3}{10}, m_2 = 0.3$ **f** $m_1 = 1\frac{1}{5}, m_2 = -\frac{6}{5}$

- 2** Find the gradient of a line that is parallel to a line with gradient:

a 4 **b** -2 **c** $\frac{1}{3}$ **d** -0.2

- 3** Find the gradient of a line that is perpendicular to a line with gradient:

a 1 **b** -6 **c** -1.5 **d** $\frac{5}{2}$

- 4** What is the gradient of a line that is perpendicular to a line with a gradient of 0.8? Select the correct answer **A**, **B**, **C** or **D**.

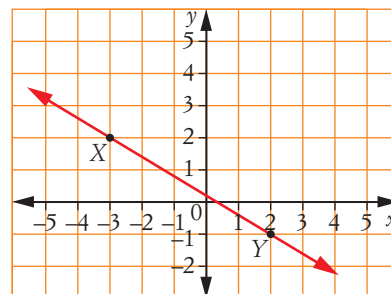
A 0.2 **B** -0.2 **C** 1.25 **D** -1.25

- 5** What is the gradient of a line that is parallel to a line that goes through $P(0, 3)$ and $Q(5, -2)$? Select **A**, **B**, **C** or **D**.

A 1 **B** -1 **C** $\frac{1}{5}$ **D** $-\frac{1}{5}$

- 6** What is the gradient of a line perpendicular to line XY shown on the right? Select **A**, **B**, $\frac{3}{5}$ or $\frac{1}{5}$.

A $\frac{5}{3}$ **B** -5 **C** $\frac{3}{5}$ **D** $\frac{1}{5}$



See Example 5

See Example 6

See Example 7

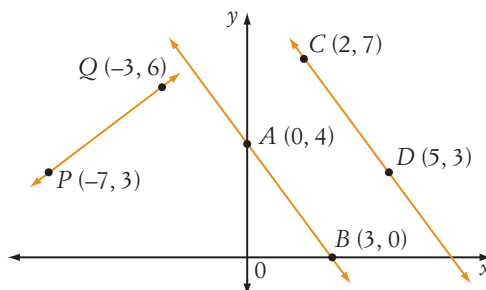
Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Coordinate geometry

- 7 Calculate the gradient of each line shown below and test whether:

a $AB \parallel CD$

b $PQ \perp CD$.



- 8 A line passes through the points $R(-5, 2)$ and $S(1, 4)$. What is the gradient of a line:

a parallel to RS ?

b perpendicular to RS ?

Skillsheet

Starting GeoGebra

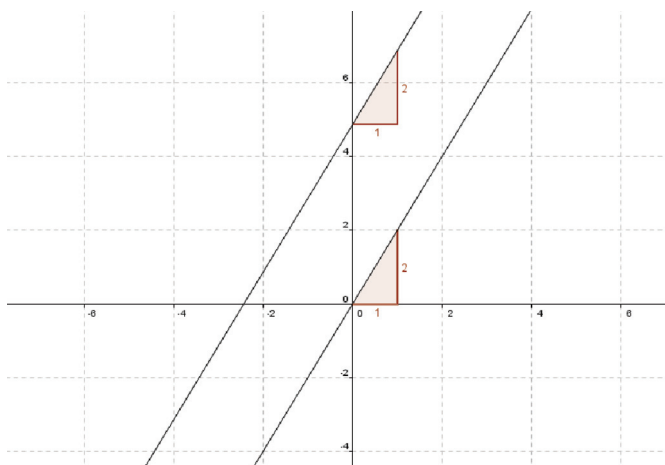
MAT10MGSS10006

Technology Parallel and perpendicular lines

This activity uses GeoGebra to find out if sets of linear equations are parallel or perpendicular.

Parallel lines

- 1 Show the **Axes and Grid**.
- 2 Use the **Input bar** to enter the pair of linear equations $y = 2x + 5$ and $y = 2x$.
- 3 Use **Move Graphics View**  and **Zoom In**  to enlarge the axes if required.
- 4 Find the **Slope** (gradient) of each line.  **Slope**
- 5 Check if the two lines are parallel, using $m_1 = m_2$



Since $m_1 = m_2 = 2$, this pair of lines is parallel.

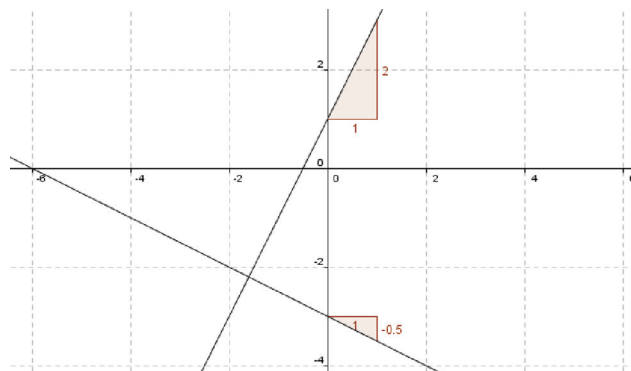
- 6 Repeat steps 1 to 5 for the pairs of equations below. Decide if the lines are parallel or not.
 - a $5x - 3y = 0$ and $y = \frac{5x}{3}$
 - b $x + y + 4 = 0$ and $x + y - 6 = 0$
 - c $x - 2y = 0$ and $y = 0.5x$
 - d $y = 5x - 9$ and $5x - y - 1 = 0$

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Perpendicular lines

- 1 Show the **Axes and Grid**.
- 2 Use the **Input bar** to enter the pair of linear equations $y = 2x + 1$ and $y = -0.5x - 3$.
- 3 Use **Move Graphics View** and **Zoom In** to enlarge the axes if required.
- 4 Find the **Slope** (gradient) of each line.
- 5 Check if the two lines are perpendicular, using $m_1 \times m_2 = -1$



Since $2 \times (-0.5) = -1$, the two lines are perpendicular.

- 6 Repeat steps 1 to 5 for the pairs of equations below. Decide if the lines are perpendicular or not.
 - a $y = 0.6x + 2$ and $y = \frac{5}{3}x$
 - b $x - 4y + 1 = 0$ and $y = -4x - 3$
 - c $3x - 2y = 0$ and $y = -\frac{2x}{3}$
 - d $y = 2x + 4$ and $x - 2y - 1 = 0$

3-03 Graphing linear equations

A relationship between two variables, x and y , whose graph is a straight line is called a **linear relationship**. The expression of that relationship as an algebraic formula, such as $y = 3x + 2$, is called a **linear equation**.

Example 8

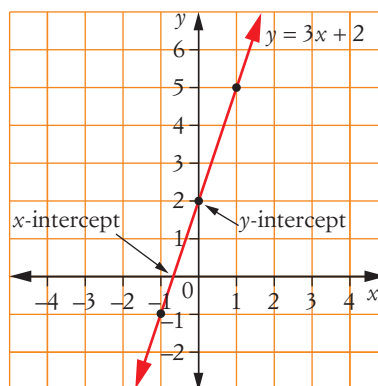
Graph $y = 3x + 2$ on a number plane.

Solution

Complete a table of values. Choose x -values close to 0 for easy calculation and graphing.

x	-1	0	1
y	-1	2	5

Graph $(-1, -1)$, $(0, 2)$ and $(1, 5)$ on a number plane. Rule a straight line through the points, place arrows at each end, and label the line with its equation.



Worksheet

Graphing linear equations

MAT10NAWK10010

Worksheet

Graphing linear equations (Advanced)

MAT10NAWK10203

Skillsheet

Graphing linear equations

MAT10NASS10005

Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Coordinate geometry

Note:

- the **x-intercept** of the line is $-\frac{2}{3}$; this is the x value where the line cuts the x -axis
- the **y-intercept** of the line is 2; this is the y value where the line cuts the y -axis
- every point on the line follows the linear equation $y = 3x + 2$. For example, $(-1, 1)$, $(0, 2)$ and $(1, 5)$ lie on the line and follow the rule $y = 3x + 2$
- there are an infinite number of points that follow the rule. Arrows on both ends of the line indicate that it has infinite length.

Stage 5.3

Using x- and y-intercepts to graph lines

We can also graph a linear equation by finding its x - and y -intercepts first.

Since any point on the x -axis has a y -coordinate of 0, we can substitute $y = 0$ into the equation to find the x -intercept.

Similarly, any point on the y -axis has an x -coordinate of 0, so we can substitute $x = 0$ into the equation to find the y -intercept.

Summary

- To find the x -intercept, substitute $y = 0$ and solve the equation.
- To find the y -intercept, substitute $x = 0$ and solve the equation.

Example 9

Find the x - and y -intercepts of the line $2x - 3y = 6$ and draw its graph.

Solution

For the x -intercept, $y = 0$.

$$2x - 3 \times 0 = 6$$

$$2x = 6$$

$$x = 3$$

The x -intercept is 3.

For the y -intercept, $x = 0$.

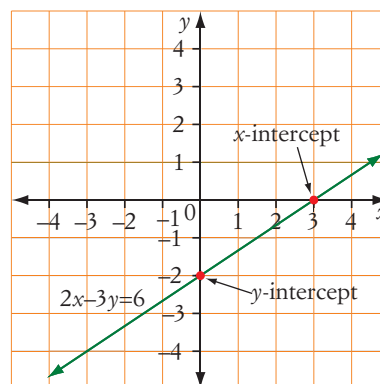
$$2 \times 0 - 3y = 6$$

$$-3y = 6$$

$$y = -2$$

The y -intercept is -2 .

Plot both intercepts on the axes, draw a line through the two points and label the line with its equation.



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Testing if a point lies on a line

Summary

A point lies on a line if its (x, y) coordinates satisfy the equation of the line.

Example 10

Which of the following points lie on the line $x - 2y = 5$?

- a** $(17, 6)$ **b** $(8, -4)$

Solution

- Separate the equation into its left-hand side (LHS) and right-hand side (RHS)
- Substitute the coordinates of the point into both sides
- If $LHS = RHS$, the point satisfies the equation and so lies on the line
- If $LHS \neq RHS$, the point does **not** lie on the line.

- a** Substitute $x = 17, y = 6$ into $x - 2y = 5$.

$$\begin{aligned} LHS &= x - 2y & RHS &= 5 \\ &= 17 - 2 \times 6 \\ &= 5 \end{aligned}$$

$LHS = RHS$, so $(17, 6)$ lies on the line.

- b** Substitute $x = 8, y = -4$ into $x - 2y = 5$.

$$\begin{aligned} LHS &= x - 2y & RHS &= 5 \\ &= 8 - 2 \times (-4) \\ &= 16 \end{aligned}$$

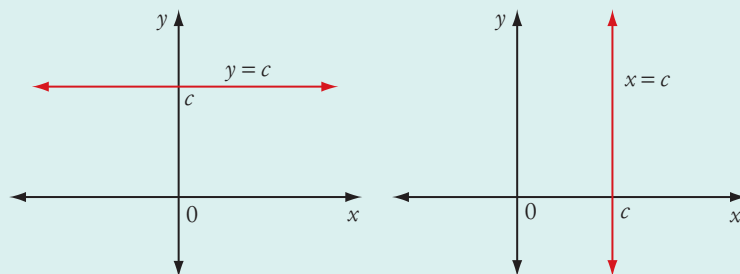
$LHS \neq RHS$, so $(8, -4)$ does not lie on the line.

Horizontal and vertical lines

Summary

The **equation of a horizontal line** is of the form $y = c$ (where c is a constant number).

The **equation of a vertical line** is of the form $x = c$ (where c is a constant number).



Technology worksheet

Horizontal and vertical
lines

MAT10NACT10001

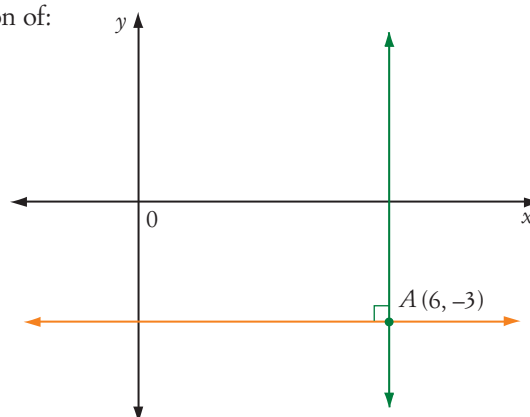
Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Coordinate geometry

Example 11

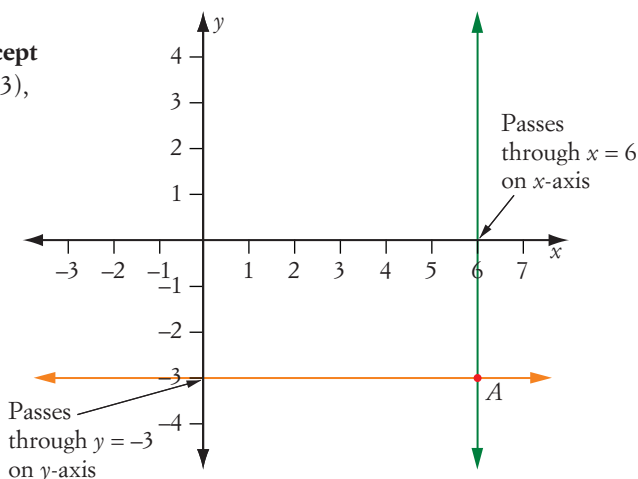
For the graph on the right, find the equation of:

- the vertical line
- the horizontal line



Solution

- The vertical line has an **x-intercept of 6** and passes through $A(6, -3)$, so its equation is $x = 6$.
- The horizontal line has a **y-intercept of -3** and passes through $A(6, -3)$, so its equation is $y = -3$.



Exercise 3-03 Graphing linear equations

See Example 7

- Graph each linear equation on a number plane, and write:

- | | | |
|-------------------|---------------------|-------------------------|
| i its x-intercept | ii its y-intercept. | |
| a $y = 3x - 1$ | b $y = 2x + 5$ | c $y = -x + 4$ |
| d $y = -2x - 2$ | e $y = 4x$ | f $y = \frac{x}{2} + 3$ |

Stage 5.3

See Example 9

- Graph each linear equation after finding its x- and y-intercepts.

- | | | |
|----------------------|----------------------|--------------------|
| a $y = 4 - 2x$ | b $2x = 4y - 8$ | c $y - x = 6$ |
| d $3x - 2y = 12$ | e $2x + 2y = 5$ | f $6 - x = 2y$ |
| g $y = 4 + 2x$ | h $5x + 3y - 15 = 0$ | i $3x - y = 6$ |
| j $2x - 5y - 20 = 0$ | k $4x + 2y - 8 = 0$ | l $x - 4y - 2 = 0$ |

See Example 10

- Test whether the point $(3, -1)$ lies on each line.

- | | | |
|----------------|---------------|--------------------|
| a $y = 2x - 5$ | b $x - y = 4$ | c $y + 2x = 5$ |
| d $y = x - 4$ | e $x + y = 5$ | f $3x + y + 8 = 0$ |

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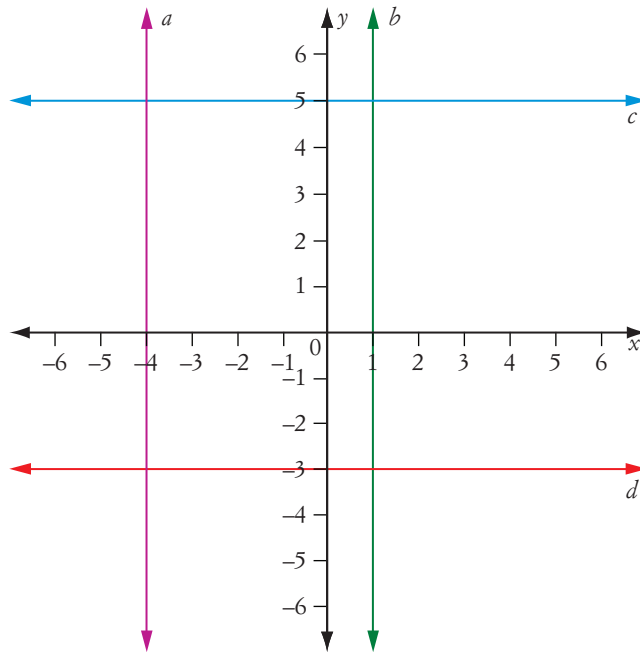
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4 Which of these points lies on the line $y = 6x - 5$? Select the correct answer **A**, **B**, **C** or **D**.

- A** $(-1, 11)$ **B** $(3, -13)$ **C** $(-2, -17)$ **D** $(-5, 25)$

5 Find the equation of each line shown below.

See Example 11



6 Graph each set of lines on a number plane.

- a** $x = 2\frac{1}{2}$, $y = -3$, $y = 1$ **b** $x = 6$, $y = -2$, $x = -\frac{1}{2}$

7 Find the equation of the line that is:

- a** horizontal and passes through the y -axis at 2
- b** vertical with an x -intercept of 4
- c** parallel to the y -axis and passes through the point $(-1, 4)$
- d** parallel to the x -axis and passes through the point $(0, -2)$
- e** 3 units above the x -axis
- f** 1 unit to the left of the y -axis
- g** drawn through the points $(-1, 6)$ and $(2, 6)$
- h** drawn through the points $(-1, 8)$ and $(-1, 2)$.

8 Which of these points lies on the line $4x + y = 1$? Select **A**, **B**, **C** or **D**.

- A** $(-1, 5)$ **B** $(-2, 7)$ **C** $(6, 9)$ **D** $(-\frac{1}{2}, 1)$

9 Which equation represents a line that is horizontal and passes through the point $(8, -2)$?
 Select **A**, **B**, **C** or **D**.

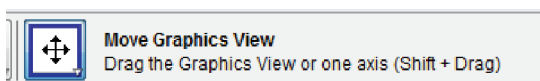
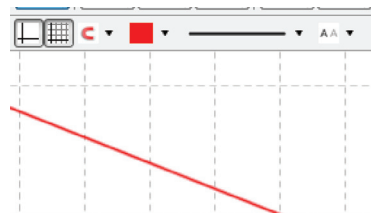
- A** $y = 8$ **B** $x = 8$ **C** $y = -2$ **D** $x = -2$

10 **a** What is another name for the line $y = 0$?

- b** What is another name for the line $x = 0$?

Technology Graphing $y = mx + b$

- 1 Show the **Axes and Grid**.
- 2 Enter the four lines $y = 3x + 2$, $y = 5x + 2$, $y = -2x + 2$, $y = -0.1x + 2$, using **Input** at the bottom of the screen.
- 3 Each straight line can be a different colour. Right-click on a line and choose a colour.
- 4 Find the **Slope** of each line.
- 5 Find the **y-intercept** of each line. Click on the right drop-down menu and use the mouse to zoom in on the y-intercept. Read off the value.



- 6 Save your GeoGebra file.
- 7 Record your results in a table as shown.

	Equation	Gradient	y-intercept
a			
b			
c			
d			

- 8 What do you notice about your results?
- 9 Repeat the steps above for each set of equations.

a $y = -4x$
 $4x + y + 1 = 0$
 $y = -4x - 10$

b $y = 2x + 3$
 $7x + y - 3 = 0$
 $0.2x - y + 3 = 0$

c $x + y + 1 = 0$
 $y = -x - 1$
 $x + y = -1$
- 10 For each set of lines drawn in question 9, complete a table as shown in Step 7 above.
- 11 What do you notice about each set of lines? Identify any key features of each set of graphs, such as gradients and y-intercepts.

NSW

Puzzle sheet

Equations in gradient form

MAT10NAPS00011

Technology worksheet

Excel spreadsheet:
Drawing linear graphs:
gradient and
y-intercept

MAT10NACT00039

3-04

The gradient–intercept equation $y = mx + b$

Summary

The equation of a straight line is $y = mx + b$, where m is the **gradient** and b is the **y-intercept**.

For this reason, $y = mx + b$ is also called the **gradient–intercept form** of a linear equation.

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Example 12

Find the gradient and y-intercept of the line with equation:

a $y = -4x + 9$ b $y = 10 - 6x$ c $y = \frac{5x + 4}{2}$ d $3x + 2y - 6 = 0$

Solution

a $y = -4x + 9$ is in the form $y = mx + b$.

$\therefore$ Gradient $m = -4$ and y-intercept $b = 9$.

b $y = 10 - 6x$ can be rewritten as $y = -6x + 10$.

$\therefore$ Gradient $m = -6$ and y-intercept $b = 10$.

c For $y = \frac{5x + 4}{2} = \frac{5x}{2} + \frac{4}{2} = \frac{5x}{2} + 2$, gradient $m = \frac{5}{2}$ and y-intercept $b = 2$

d $3x + 2y - 6 = 0$ can be rearranged in the form $y = mx + b$.

$$3x + 2y - 6 - 3x = 0 - 3x$$

$$2y - 6 = -3x$$

$$2y - 6 + 6 = -3x + 6$$

$$2y = -3x + 6$$

$$y = \frac{-3x + 6}{2}$$

$$y = \frac{-3x}{2} + 3$$

$\therefore 3x + 2y - 6 = 0$ has gradient $m = -\frac{3}{2}$ and y-intercept $b = 3$.

Video tutorial

The gradient-intercept formula

MAT10NAVT10011

Example 13

Graph each linear equation by finding the gradient and y-intercept first.

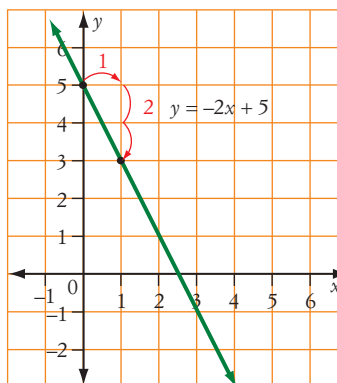
a $y = -2x + 5$ b $y = \frac{3}{4}x - 2$

Solution

a $y = -2x + 5$ has a gradient of -2 and a y-intercept of 5 .

- Plot the y-intercept 5 on the y-axis.
- Make a gradient of -2 by moving **across** 1 unit (**run**) and **down** 2 units (**'negative' rise**) and marking the point at $(1, 3)$.
- Rule a line through this point and the y-intercept.

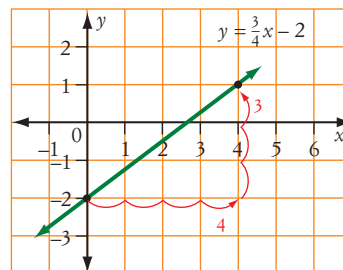
Don't forget to label the line with its equation ' $y = -2x + 5$ '



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Coordinate geometry

- b $y = \frac{3}{4}x - 2$ has a gradient of $\frac{3}{4}$ and a y-intercept of -2 .
- Plot the y-intercept -2 on the y-axis.
 - Make a gradient of $\frac{3}{4}$ by moving **across** 4 units (**run**) and **up** 3 units (**rise**) and marking the point at $(4, 1)$.
 - Rule a line through this point and the y-intercept.



Example 14

Which of the following lines is parallel to $y = -2x + 3$?

- A $y = 2x + 3$ B $y = -2x + 1$ C $y = -2x$ D $y = 5x + 3$

Solution

Parallel lines have the **same gradient**. The line $y = -2x + 3$ has the gradient $m = -2$.

- A $y = 2x + 3$ has gradient 2.
- B $y = -2x + 1$ has gradient -2 .
- C $y = -2x$ has gradient -2 .
- D $y = 5x + 3$ has gradient 5.

$\therefore$ The lines **B** ($y = -2x + 1$) and **C** ($y = -2x$) are parallel to $y = -2x + 3$.

Exercise 3-04 The gradient–intercept formula $y = mx + b$

See Example 12

- 1 Find the gradient and y-intercept of each line below.

- a $y = 3x - 2$ b $y = -2x + 7$ c $y = x + 4$ d $y = 9 - x$
e $y = \frac{3x}{4} + 6$ f $y = x$ g $y = \frac{x}{2} - 11$ h $y = \frac{2x + 18}{3}$
i $y = \frac{-24 - x}{3}$ j $y = 2(x - 3)$ k $11 - 3x = y$ l $\frac{2x - 7}{2} = y$

- 2 Find the equation of a line with:

- a a gradient of 2 and a y-intercept of 1 b a gradient of $\frac{3}{4}$ and a y-intercept of 2
c a gradient of -7 and a y-intercept of 5 d a gradient of $-\frac{2}{5}$ and a y-intercept of 3
e $m = -2, b = -3$ f $m = -3, b = \frac{1}{2}$

See Example 13

- 3 Graph each linear equation by finding the gradient and y-intercept first.

- a $y = 2x + 1$ b $y = 3x - 2$ c $y = 2x$ d $y = \frac{x}{2} - 1$
e $y = -2x + 3$ f $y = -\frac{3x}{4}$ g $y = \frac{-5x + 2}{2}$ h $y = \frac{3x - 20}{5}$

- 4 Write the equation of a line with a gradient of 2 and a y-intercept of 0.

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- 5 Select the lines that are parallel to the given line each time. There may be more than one answer.

See Example 14

a $y = x + 6$

A $y = 6x$

B $y = 6 - x$

C $y = x + 1$

D $y = 2x$

b $y = 3x + 10$

A $y = 10x + 3$

B $y = 3x - 1$

C $y = 1 - 3x$

D $y = 4 + 3x$

c $y = \frac{x}{2} + 5$

A $y = 2x - 1$

B $y = \frac{x+6}{2}$

C $y = 1 - \frac{x}{2}$

D $y = x + 2$

d $y = 6$

A $y = 2x + 6$

B $y = 6x$

C $y = -1$

D $y = 10$

e $y = 4x$

A $y = 4x - 2$

B $y = 4x + 3$

C $y = 4$

D $y = 1 - 4x$

f $x = 10$

A $y = 10$

B $y = 10x$

C $x = 2y$

D $x = -6$

- 6 For each set of linear equations, find a pair of equations whose graphs are parallel lines.

a $y = 4x + 3$ $y = x + 2$ $y = 4x - 6$ $y = 2x$

b $y = 5x + 1$ $3x - y + 7 = 0$ $y = 3x - 2$ $y = -5x + 2$

Mental skills 3

Maths without calculators

Time differences

- 1 Study each example.

- a What is the time difference between 11:40 a.m. and 6:15 p.m.?

From 11:40 a.m. to 5:40 p.m. = 6 hours

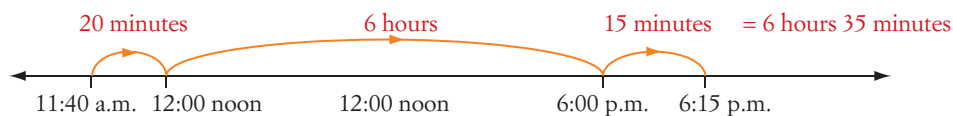
Count: '11:40, 12:40, 1:40, 2:40, 3:40, 4:40, 5:40'

From 5:40 a.m. to 6:00 p.m. = 20 min

From 6:00 p.m. to 6:15 p.m. = 15 min

5 hours + 20 min + 15 min = 6 hours 35 min

OR:



- b What is the time difference between 2030 and 0120?

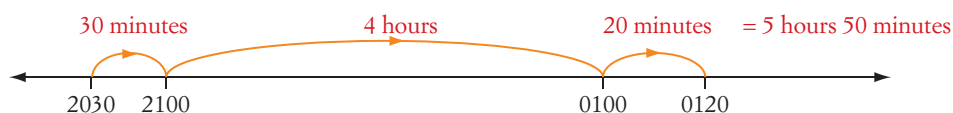
From 2030 to 0030 = 4 hours ($24 - 20 = 4$)

From 0030 to 0100 = 30 min

From 0100 to 0120 = 20 min

4 hours + 30 minutes + 20 minutes = 5 hours 50 minutes

OR:



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Coordinate geometry

2 Now find the time difference between:

a 11:10 a.m. and 7:40 p.m.

c 4:45 p.m. and 8:10 p.m.

e 1:05 p.m. and 12:30 a.m.

g 0425 and 0935

i 7:55 a.m. and 3:50 p.m.

b 6:20 pm. and 12:00 midnight

d 2:35 a.m. and 10:50 a.m.

f 9:35 a.m. and 11:15 a.m.

h 1440 and 2025

j 2:40 p.m. and 10:20 p.m.

NSW

3-05

The general form of a linear equation $ax + by + c = 0$

Puzzle sheet

Linear equations code
puzzle

MAT10NAPS10011

Worksheet

Parallel and
perpendicular lines

MAT10NAWK00015

A linear equation written in **gradient-intercept form**, such as $y = -\frac{3}{4}x + 2$, can also be written in **general form** $3x + 4y - 8 = 0$. Note that, for the general form $ax + by + c = 0$, all of the terms on the left-hand side of the equation are written with no fractions, and only 0 is on the right-hand side. Sometimes the general form is neater and more convenient.

Example 15

Write each linear equation in general form.

a $y = 6x + 2$

b $y = -\frac{2}{3}x + 2$

c $y = 2x - \frac{3}{5}$

Solution

a $y = 6x + 2$

$$0 = 6x - y + 2$$

$$6x - y + 2 = 0$$

Subtracting y from both sides.

Swapping sides so that zero appears on the RHS.

b $y = -\frac{2}{3}x + 2$

$$3y = 3\left(-\frac{2}{3}x + 2\right)$$

$$= -2x + 6$$

$$2x + 3y = 6$$

$$2x + 3y - 6 = 0$$

Multiplying both sides by 3 to remove the fraction.

Adding $2x$ to both sides.

Subtracting 6 from both sides.

c $y = 2x - \frac{3}{5}$

$$5y = 5\left(2x - \frac{3}{5}\right)$$

$$5y = 10x - 3$$

$$0 = 10x - 5y - 3$$

$$10x - 5y - 3 = 0$$

Multiplying both sides by 5 to remove the fraction.

Subtracting $5y$ from both sides.

Swapping sides so that zero appears on the RHS.

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Summary

The **general form of a linear equation** is written as $ax + by + c = 0$, where a , b and c are integers and a is positive.

Example 16

Find the gradient and y -intercept of the line whose equation is $5x + 2y - 10 = 0$.

Solution

Rewrite $5x + 2y - 10 = 0$ in the form $y = mx + b$.

$$5x + 2y - 10 = 0$$

$$2y - 10 = -5x$$

$$2y = -5x + 10$$

$$\frac{2y}{2} = \frac{-5x + 10}{2}$$

$$y = \frac{-5x}{2} + 5$$

$$\therefore \text{Gradient: } m = -\frac{5}{2}, \text{ } y\text{-intercept: } b = 5$$

Aim to have y on its own on the LHS of the equation.

Subtracting $5x$ from both sides.

Adding 10 to both sides.

Dividing both sides by 2.

Exercise 3-05

The general form of a linear equation $ax + by + c = 0$

- 1 Write each linear equation in general form.

a $y = x + 2$

b $y = 3x - 1$

c $y = 8 + 5x$

d $x + 2y = 3$

e $x - 2y = 6$

f $y = 8x + 2$

g $y + 3 = 6x$

h $2y = x - 6$

i $y = \frac{3}{5}x + 2$

- 2 Find the gradient and y -intercept of the line with each equation.

a $2x + y = 6$

b $8x - 2y = 10$

c $3x - 2y + 4 = 0$

d $y + 2x - 1 = 0$

e $2x + y + 5 = 0$

f $4x + 3y - 12 = 0$

- 3 Find the gradient, m , and the y -intercept, b , of the line with equation $x - 3y + 5 = 0$.
 Select the correct answer **A**, **B**, **C** or **D**.

A $m = -1, b = 5$

B $m = \frac{1}{3}, b = \frac{5}{3}$

C $m = 1, b = -5$

D $m = \frac{1}{3}, b = -\frac{5}{3}$

- 4 Which statement is false about the line whose equation is $3x + y - 6 = 0$? Select **A**, **B**, **C** or **D**.

A The gradient is -3 .

B The y -intercept is -6 .

C The x -intercept is 2.

D It is parallel to the line $y = -3x$.

See Example 15

See Example 16

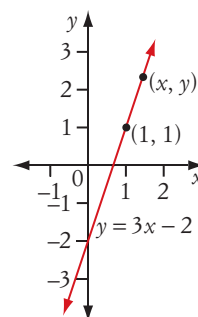
Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Coordinate geometry

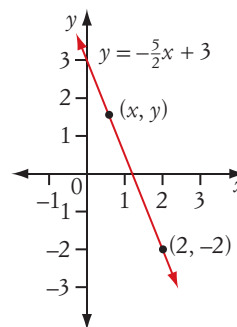
Stage 5.3

Investigation: The equation of a line given its gradient and a point

- 1 The graph shows the line $y = 3x - 2$.
 - a What is its gradient?
 - b If (x, y) is any other point on the line, show that $m = \frac{y-1}{x-1}$.
 - c Explain why $\frac{y-1}{x-1} = 3$
 - d Hence show that $y - 1 = 3(x - 1)$ and simplify this equation to obtain $y = 3x - 2$.



- 2 The graph shows the line $y = -\frac{5}{2}x + 3$.
 - a What is its gradient?
 - b If (x, y) is any other point on the line, show that $m = \frac{y+2}{x-2}$.
 - c Explain why $\frac{y+2}{x-2} = -\frac{5}{2}$
 - d Hence show that $y + 2 = -\frac{5}{2}(x - 2)$ and simplify this expression to obtain $y = -\frac{5}{2}x + 3$.



- 3 The equation of a line is given by $\frac{y-2}{x-7} = \frac{3}{4}$.
 - a What is the gradient of the line?
 - b Can you give the coordinates of a point on this line by looking at its equation? Why?
- 4 Write the equation of a line which passes through the point $(-3, 5)$ and has a gradient equal to 2. Compare your result with other groups.
- 5 A line with gradient m passes through the point (x_1, y_1) .
 - a Show that $m = \frac{y-y_1}{x-x_1}$, where (x, y) is any other point on the line.
 - b Explain why $y - y_1 = m(x - x_1)$.

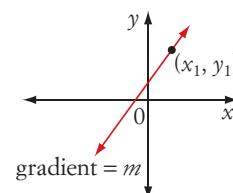
NSW

3-06

The point-gradient form of a linear equation

There is a formula for finding the equation of a line if we know its gradient m and a point on the line (x_1, y_1) . Let (x, y) be any other point on the line.

Then $m = \frac{y-y_1}{x-x_1}$ or $y - y_1 = m(x - x_1)$.



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Summary

The equation of a line with gradient m and which passes through the point (x_1, y_1) is:

$$y - y_1 = m(x - x_1)$$

It is called the **point-gradient form of a linear equation**.

Example 17

Find the equation of the line with a gradient of $\frac{2}{3}$ that passes through the point $(-2, 1)$.

Solution

$$m = \frac{2}{3}, x_1 = -2, y_1 = 1.$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{2}{3}[x - (-2)]$$

$$3(y - 1) = 2(x + 2)$$

$$3y - 3 = 2x + 4$$

$$0 = 2x - 3y + 7$$

$$2x - 3y + 7 = 0$$

In general form

Example 18

Find the equation of the line passing through the points $(1, 3)$ and $(4, -3)$.

Solution

First find the gradient of the line by using the points $(1, 3)$ and $(4, -3)$.

$$m = \frac{-3 - 3}{4 - 1}$$

$$= \frac{-6}{3}$$

$$= -2$$

Now use $y - y_1 = m(x - x_1)$ with $m = -2$ and $(1, 3)$.

$$y - 3 = -2(x - 1)$$

$$= -2x + 2$$

$$y = -2x + 5 \quad \text{or} \quad 2x + y - 5 = 0 \text{ in general form}$$

OR: Using the other point $(4, -3)$ instead:

$$y - (-3) = -2(x - 4)$$

$$y + 3 = -2x + 8$$

$$y = -2x + 5 \quad \text{or} \quad 2x + y - 5 = 0 \text{ in general form}$$

Either of the points $(1, 3)$ or $(4, -3)$ can be used to find the equation of the line.

Stage 5.3

Video tutorial

The point-gradient formula

MAT10NAVT10012

Stage 5.3

See Example 17

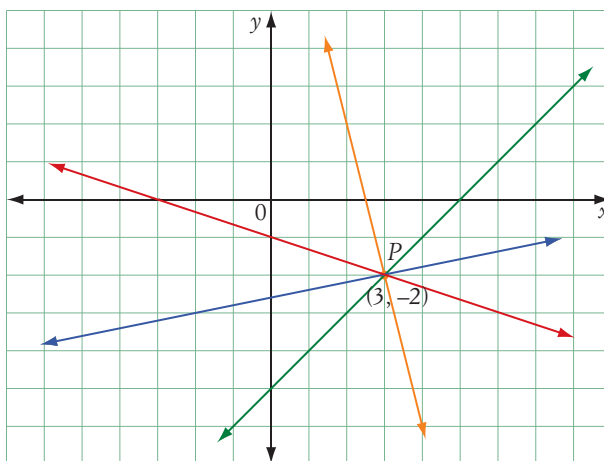
Exercise 3-06

The point–gradient form of a linear equation

In this exercise, express all equations of lines in general form.

- Find the equation of each line, given a point on the line and the gradient.

a (2, 5), gradient 2	b (−6, 4), gradient −1	c (3, −8), gradient 4
d (−1, −2), gradient $\frac{2}{3}$	e (2, −8), gradient $-\frac{1}{5}$	f (−1, 7), gradient −3
g $(\frac{1}{2}, -3)$, gradient −4	h $(-4, -\frac{1}{2})$, gradient $\frac{3}{4}$	i (−2, −6), gradient −2
- Four lines a , b , c and d intersect at $P(3, -2)$. The gradients of a , b , c and d are 1, $-\frac{1}{3}$, −4 and $\frac{1}{5}$ respectively.



- Copy the diagram and correctly label the lines a , b , c and d .
- Find the equation of each line.

See Example 18

- Find the equation of the line passing through each pair of points.

a (7, 3) and (10, 6)	b (8, 10) and (−2, 2)	c (−1, 3) and (5, 8)
d (2, −2) and (−1, 6)	e (4, −3) and (6, −6)	f (−1, −2) and (2, 3)
g (−10, 2) and (1, −4)	h (−3, 6) and (1, 2)	i (−4, −9) and (−1, −5)
- Two lines, k and l , intersect at (−1, 4). Line k has a gradient of $-\frac{1}{2}$, while line l has a gradient of 3. Find the equations of lines k and l .
- Find the equation of a line with a gradient of −4 and an x -intercept of 5.
- A line passes through the y -axis at (0, 6) and has a gradient of $\frac{5}{7}$. What is its equation?
- A line with a gradient of $\frac{2}{3}$ passes through the midpoint of (5, 6) and (1, 10). Find the equation of the line.
- A line with a gradient of $-\frac{3}{5}$ passes through the midpoint of (−8, −2) and (−2, 20). Find its equation.

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- 9 a The gradient–intercept form of a line, $y = mx + b$, can also be used to find the equation of a line given its gradient and a point on the line. Use $y = mx + b$ to find the equation of the line with gradient 2 that passes through the point (2, 5).
- b Compare your equation with your answer to question 1a.
- 10 a The point–gradient formula can be converted to a formula for finding the equation of a line passing through two points (x_1, y_1) and (x_2, y_2) . Prove that the ‘two-point formula’ is $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$.
- b Use the two-point form to find the equation of a line passing through the points (7, 3) and (10, 6).
- c Compare your equation with your answer to question 3a.

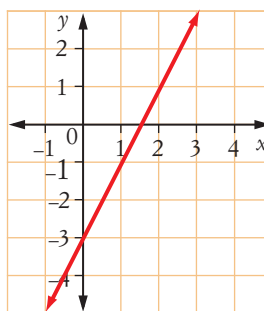
Stage 5.3

3-07 Finding the equation of a line

NSW

Example 19

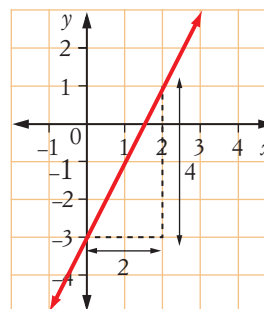
Find the equation of the line.



Solution

Select two points on the line to find the gradient, say (0, -3) and (2, 1).

$$\text{Gradient } m = \frac{\text{rise}}{\text{run}} = \frac{4}{2} = 2$$



y-intercept: $b = -3$

$\therefore$ The equation of the line is $y = 2x - 3$.

We can check that this equation is correct for any point on the line, say (3, 3).

When $x = 3$, $y = 2 \times 3 - 3 = 3$.

from the graph

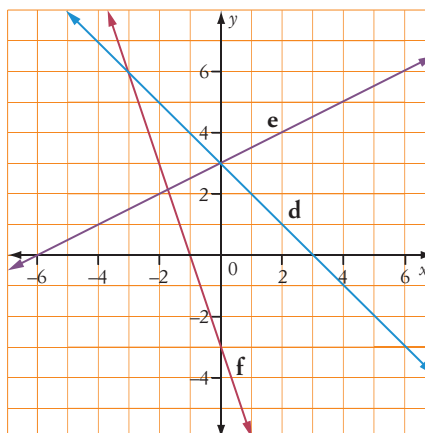
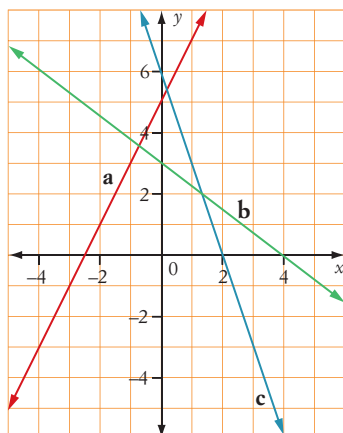
$$y = mx + b$$

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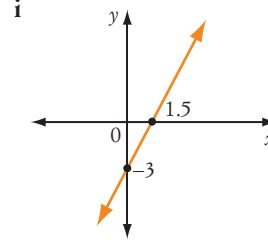
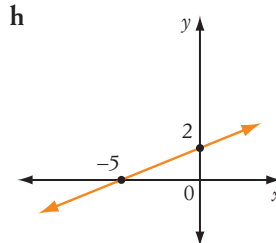
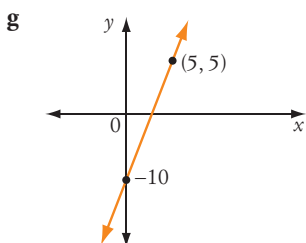
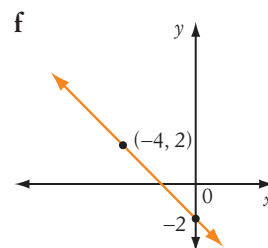
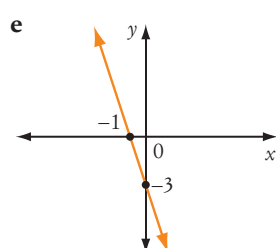
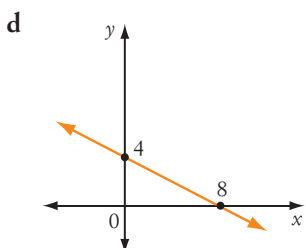
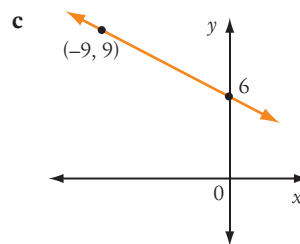
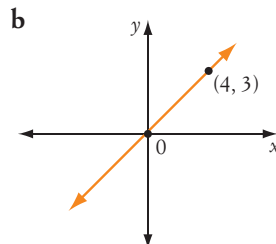
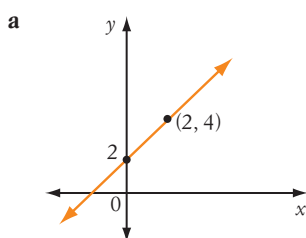
Coordinate geometry

Exercise 3-07 Finding the equation of a line

See Example 19 1 Find the equation of each line.



2 Find the equation of each line.



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Investigation: Sausage sizzle

A local football club is organising a sausage sizzle on Saturday to raise money to buy new equipment. It costs \$25 to hire a gas bottle to run the barbecue and each sandwich costs \$0.90 to make.



Fairfax Syndication/Craig Abraham

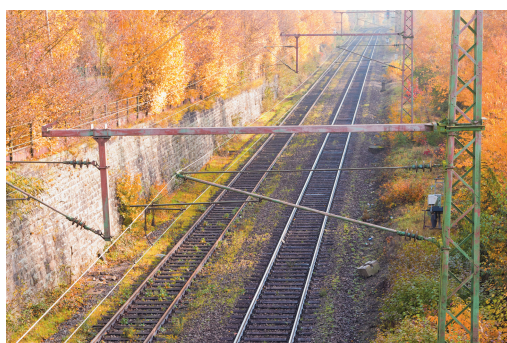
- Copy and complete this table below to show the cost of making sausage sandwiches. Include the cost of hiring the gas bottle.

No. of sandwiches (x)	0	10	20	30	40	50	60	70	80	90	100
Cost (\$ y)	25	34									

- Find the linear equation (formula) for y that represents the cost of making x sausage sandwiches.
- Use an appropriate scale to construct a graph that shows the cost of making from $x = 0$ to $x = 100$ sandwiches. Label your axes and give your graph an appropriate title.
- How much will it cost to make 35 sausage sandwiches?
- How many sandwiches can be made for \$98.80?
- How much would it cost to make 120 sausage sandwiches?
- If the club sold 75 sausage sandwiches for \$3 each, how much money would they take?
 - How much **profit** would the club make?

3-08

Equations of parallel and perpendicular lines



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Summary

If two lines with gradients m_1 and m_2 are parallel, then $m_1 = m_2$.

If two lines with gradients m_1 and m_2 are perpendicular, then $m_1 \times m_2 = -1$ or $m_2 = -\frac{1}{m_1}$.

Puzzle sheet

Linear equations
match-up

MAT10NAPS10012

Worksheet

Writing equations
of lines

MAT10NAWK10013

Puzzle sheet

Equations of
parallel lines

MAT10NAPS00013

Technology

GeoGebra:
Perpendicular lines

MAT10NATC00005

Video tutorial

Coordinate geometry

MAT10NAVT00005

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Coordinate geometry

Example 20

Find the equation of the line parallel to $y = 8 - 3x$ that passes through the point $(-1, 6)$.

Solution

For $y = 8 - 3x$ (or $y = -3x + 8$), the gradient is $m = -3$.

A line parallel to $y = 8 - 3x$, will also have $m = -3$.

Using the point–gradient formula $y - y_1 = m(x - x_1)$ with $m = -3$ and point $(-1, 6)$:

$$\begin{aligned} y - 6 &= -3[x - (-1)] \\ &= -3(x + 1) \\ &= -3x - 3 \\ y &= -3x + 3 \end{aligned}$$

OR: Using the gradient–intercept equation $y = mx + b$:

$$y = -3x + b$$

To find the value of b , substitute the point $(-1, 6)$ into the equation:

$$\begin{aligned} y &= -3x + b \\ 6 &= -3 \times (-1) + b & x = -1, y = 6 \\ 6 &= 3 + b \\ b &= 3 \end{aligned}$$

$\therefore$ The equation is $y = -3x + 3$.

Example 21

Find the equation of the line perpendicular to $3x - 4y + 6 = 0$, which passes through the point $(5, 4)$.

Solution

To find the gradient of $3x - 4y + 6 = 0$, first convert it to the form $y = mx + b$:

$$\begin{aligned} 3x - 4y + 6 &= 0 \\ 3x + 6 &= 4y \\ 4y &= 3x + 6 \\ y &= \frac{3x + 6}{4} \\ y &= \frac{3}{4}x + \frac{3}{2} \end{aligned}$$

$$y = mx + b$$

$$\therefore \text{Gradient} = \frac{3}{4}$$

$$\begin{aligned} \therefore \text{Gradient of perpendicular line} &= \frac{-1}{\left(\frac{3}{4}\right)} \\ &= -\frac{4}{3} \end{aligned}$$

The negative reciprocal of $\frac{3}{4}$.

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Using the point-gradient formula $y - y_1 = m(x - x_1)$

with $m = -\frac{4}{3}$ and point (5, 4):

$$y - 4 = -\frac{4}{3}(x - 5)$$

In general form

$$3(y - 4) = -4(x - 5)$$

$$3y - 12 = -4x + 20$$

$$4x + 3y - 32 = 0$$

OR: Using the gradient-intercept equation $y = mx + b$:

$$y = -\frac{4x}{3} + b$$

To find the value of b , substitute the point (5, 4) into the equation.

$$x = 5, y = 4$$

$$4 = \left(-\frac{4}{3}\right) \times 5 + b$$

$$= -\frac{20}{3} + b$$

$$4 + \frac{20}{3} = b$$

$$b = \frac{32}{3}$$

$$\therefore \text{The equation is } y = -\frac{4x}{3} + \frac{32}{3} \text{ or } y = \frac{-4x + 32}{3}$$

or, converting to the neater general form:

$$3y = -4x + 32$$

$$4x + 3y - 32 = 0$$

Stage 5.3

Exercise 3-08 Equations of parallel and perpendicular lines

- 1 Find the equation of the line that is parallel to:
 - a $y = 2x + 9$ and has a y -intercept of 4
 - b $y = 3x$ and has a x -intercept of -2
 - c $y = 5 - \frac{x}{2}$ and passes through $(-1, 6)$
 - d $2x - y = 6$ and passes through $(5, -2)$
 - e $y = -5x - 8$ and passes through the midpoint of $(3, -10)$ and $(-5, -6)$
 - f $2y = x - 3$ and passes through $(6, -7)$
- 2 Find the equation of a line that is perpendicular to:
 - a $y = \frac{x}{2}$ and has a y -intercept of -2
 - b $y = -5x$ and has a x -intercept of 1
 - c $y = 3x - 1$ and passes through the x -axis at 4
 - d $y = \frac{x-6}{3}$ and passes through $(1, -6)$
 - e $x + y - 6 = 0$ and passes through $(-4, 2)$
 - f $3x - y - 9 = 0$ and passes through $(-10, -7)$

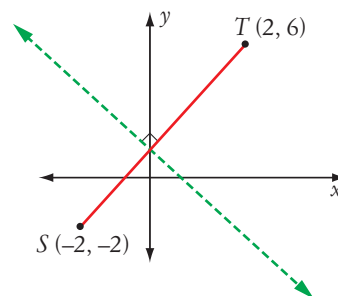
See Example 20

See Example 21

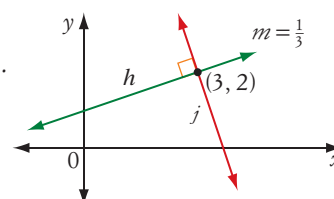
Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Coordinate geometry

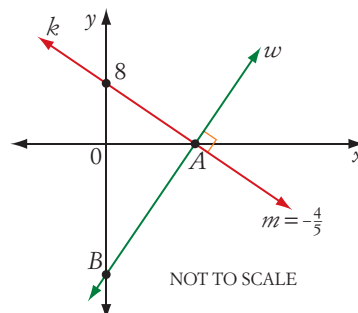
- 3 a Find the gradient of interval ST in the diagram on the right.
b Find the midpoint of ST .
c The dotted line is perpendicular to ST and passes through its midpoint. What is its gradient?
d Find the equation of the dotted line, in the form $y = mx + b$.



- 4 a Find the equation of line h in the diagram.
b Find the gradient of line j (which is perpendicular to line h).
c Find the equation of line j .



- 5 a Find the equation of line k .
b Find the coordinates of point A .
c Find the gradient of line w .
d Find the equation of line w .
e Find the coordinates of point B .



Stage 5.3
NSW

Worksheet

Geometry problems
using coordinates

MAT10NAWK10204

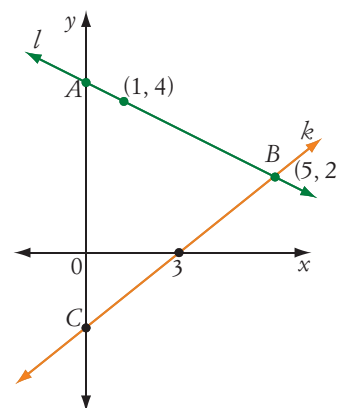
3-09 Coordinate geometry problems

A variety of problems can be solved by applying coordinate geometry methods, including proving geometric properties of triangles and quadrilaterals.

Example 22

Lines k and l are shown in the diagram. Find:

- a the equation of line k
b the equation of line l
c the coordinates of point A
d the coordinates of point C
e the area of the triangle ABC



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Solution

- a** Line k passes through $(3, 0)$ and $(5, 2)$.

$$m = \frac{2 - 0}{5 - 3} = \frac{2}{2} = 1$$

Using the point–gradient formula $y - y_1 = m(x - x_1)$:

$$y - 2 = 1(x - 5)$$

$$= x - 5$$

$$y = x - 3$$

(or $x - y - 3 = 0$ in general form)

using the point $(5, 2)$

- b** Line l passes through $(1, 4)$ and $(5, 2)$.

$$\therefore m = \frac{2 - 4}{5 - 1} = \frac{-2}{4} = -\frac{1}{2}$$

$$y - 4 = -\frac{1}{2}(x - 1)$$

$$2(y - 4) = -1(x - 1)$$

$$2y - 8 = -x + 1$$

$$x + 2y - 9 = 0$$

using the point $(1, 4)$

- c** A is the y -intercept of line l .

Substitute $x = 0$ into $x + 2y - 9 = 0$.

$$0 + 2y - 9 = 0$$

$$2y = 9$$

$$y = \frac{9}{2}$$

$$= 4.5$$

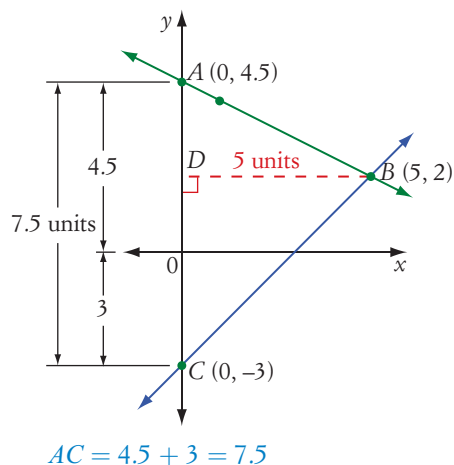
$\therefore A$ is $(0, 4.5)$

- d** C is the y -intercept of line k .

The y -intercept of $y = x - 3$ is -3 .

$\therefore C$ is $(0, -3)$

- e** Area of $\triangle ABC = \frac{1}{2} \times \text{base} \times \text{height}$
 $= \frac{1}{2} \times AC \times BD$
 $= \frac{1}{2} \times 7.5 \times 5$
 $= 18.75 \text{ units}^2$



Stage 5.3

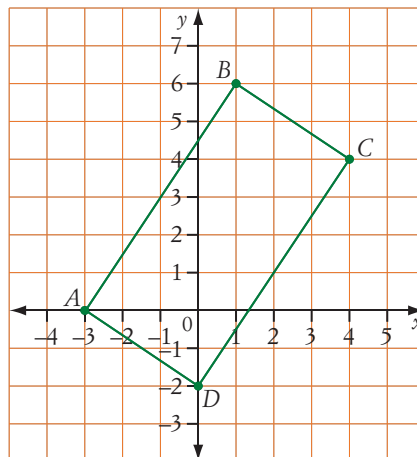
Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Coordinate geometry

Stage 5.3

Example 23

$A(-3, 0)$, $B(1, 6)$, $C(4, 4)$ and $D(0, -2)$ are the vertices of a rectangle.

- By finding the lengths of AC and BD , show that the diagonals of the rectangle are equal.
- Find the midpoints of the diagonals AC and BD .
- Show that the diagonals of the rectangle bisect each other.



Solution

- a $A(-3, 0)$, $C(4, 4)$

$$\begin{aligned} AC &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{[4 - (-3)]^2 + (4 - 0)^2} \\ &= \sqrt{65} \end{aligned}$$

- $B(1, 6)$, $D(0, -2)$

$$\begin{aligned} BD &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(0 - 1)^2 + (-2 - 6)^2} \\ &= \sqrt{65} \end{aligned}$$

$$\therefore AC = BD$$

$\therefore$ The diagonals are equal.

- b Midpoint of $AC \equiv \left(\frac{-3 + 4}{2}, \frac{0 + 4}{2} \right)$
 $\equiv \left(\frac{1}{2}, 2 \right)$

$$\begin{aligned} \text{Midpoint of } BD &\equiv \left(\frac{1 + 0}{2}, \frac{6 + (-2)}{2} \right) \\ &\equiv \left(\frac{1}{2}, 2 \right) \end{aligned}$$

- c The midpoints of both diagonals are the same, so the diagonals bisect each other.

NEW CENTURY MATHS ADVANCED 10 + 10A

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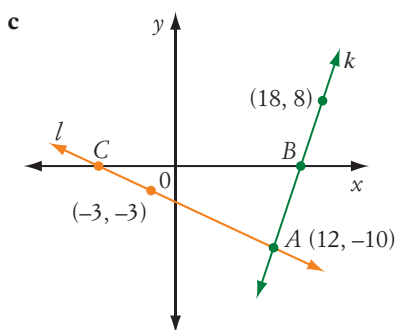
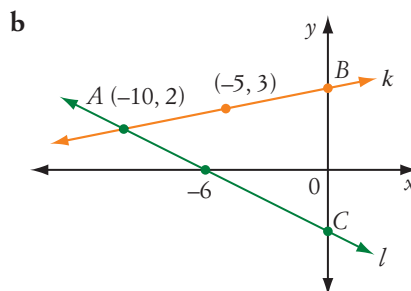
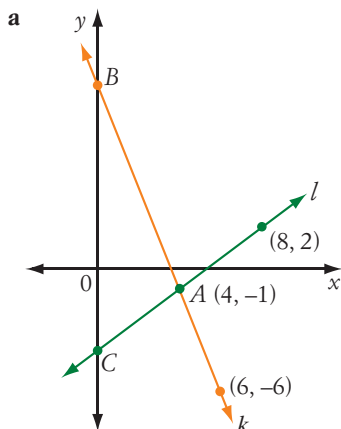
Exercise 3-09 Coordinate geometry problems

Stage 5.3

See Example 22

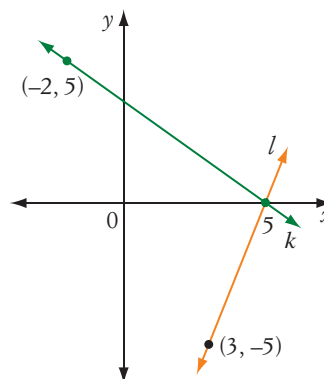
1 For each graph, find:

- i the equation of line k
- ii the equation of line l
- iii the coordinates of point B
- iv the coordinates of point C
- v the area of $\triangle ABC$



2 For this graph, find:

- a the equation of line l
- b the equation of line k
- c w if the point $(7, w)$ lies on l
- d t if the point $(t, 11)$ lies on k



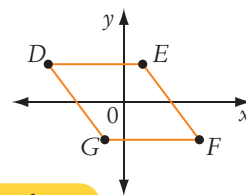
Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Coordinate geometry

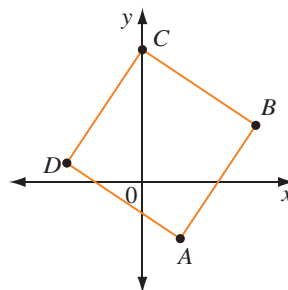
Stage 5.3

See Example 23

- 3 The vertices of a rhombus are $D(-4, 2)$, $E(1, 2)$, $F(4, -2)$ and $G(-1, -2)$.
 - a Show that all sides of the rhombus are equal.
 - b By finding their gradients, show that the opposite sides of the rhombus are parallel.
 - c Show that the diagonals DF and GE of the rhombus cross at right angles.
 - d Find the midpoints of the diagonals DF and GE . Do the diagonals bisect each other? Give reasons.
 - e List the properties of a rhombus that have been demonstrated in this question.
- 4 A quadrilateral has vertices $P(-7, 2)$, $Q(2, -7)$, $R(5, -4)$ and $S(-4, 5)$.
 - a Draw a diagram showing the given information.
 - b Find the lengths of PR and QS in surd form.
 - c Find the midpoints of PR and QS .
 - d Is PR perpendicular to QS ? Why?
 - e What type of quadrilateral is $PQRS$? Explain.
- 5 A quadrilateral has vertices $C(2, 6)$, $D(-5, 2)$, $E(-1, -5)$ and $F(6, -1)$.
 - a Draw a diagram showing the given information.
 - b Find the length of each diagonal.
 - c Find the midpoint of each diagonal.
 - d Show that the diagonals are perpendicular.
 - e What type of quadrilateral is $CDEF$? Explain.
- 6 A quadrilateral has vertices $B(1, 7)$, $C(-5, 2)$, $D(2, -2)$ and $E(8, 3)$.
 - a Find the length of each side.
 - b Find the gradient of each side.
 - c Find the midpoint of each diagonal.
 - d What type of quadrilateral is $BCDE$?
- 7 A square has vertices $A(2, -3)$, $B(6, 3)$, $C(0, 7)$ and $D(-4, 1)$.
 - a Show that its diagonals are equal.
 - b Show that its diagonals bisect each other at right angles.
 - c Hence explain why $ABCD$ is a square.
- 8 Show that the diagonals of the rhombus with vertices $K(4, 2)$, $L(-1, 4)$, $M(1, -1)$ and $N(6, -3)$ bisect each other at right angles.
- 9 Show that $W(-5, -4)$, $X(4, -1)$, $Y(6, 6)$ and $Z(-3, 3)$ are the vertices of a parallelogram by finding the gradients of each side and showing that the opposite sides are parallel.



'Show that' means all working out must be provided to fully explain your answer



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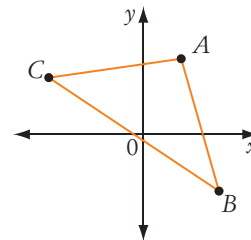
Stage 5.3

10 $J(-3, 0)$, $K(3, -2)$, $L(1, 3)$ and $M(-5, 5)$ are the vertices of a quadrilateral.

- Find the gradient of each side.
- What type of quadrilateral is $JKLM$? Explain.

11 $A(2, 4)$, $B(4, -3)$ and $C(-5, 3)$ are the vertices of a triangle.

- X and Y are the midpoints of AB and AC respectively.
Find the coordinates of X and Y .
- Find the gradients of XY and CB . Is it true that $XY \parallel CB$?
- Find the lengths of XY and CB and, hence, show that $CB = 2XY$.



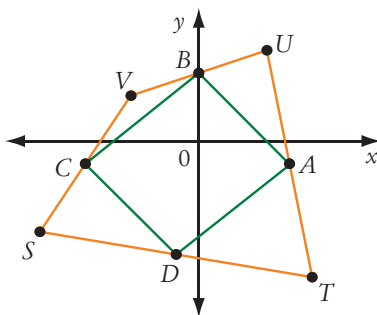
12 $C(-7, -6)$, $N(1, -3)$, $T(4, 5)$ and $W(-4, 2)$ are the vertices of a quadrilateral.

- Show that its diagonals bisect each other at right angles.
- Hence, what type of quadrilateral is $CNTW$?

13 What type of a quadrilateral is formed by the points $H(-6, 2)$, $I(6, -4)$, $J(4, 2)$ and $K(-2, 5)$?

14 Show that the points $S(5, -6)$, $T(6, 0)$, $W(-6, 2)$ and $X(-7, -4)$ are the vertices of a rectangle.

15 The points $T(5, -6)$, $U(3, 4)$, $V(-3, 2)$ and $S(-7, -4)$ are the vertices of a quadrilateral.

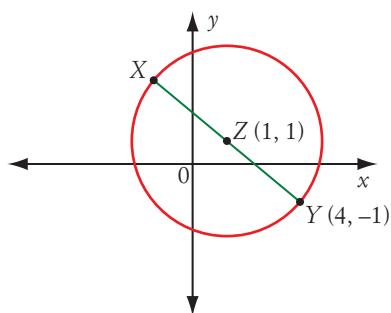


- Find the coordinates of the midpoint of each side.
 - What type of quadrilateral is formed when the midpoints of the sides are joined? Explain.
- 16 For the points $L(-8, -1)$, $M(1, 2)$ and $N(10, 5)$:
- Find the gradients of LM , LN and MN .
 - What can you say about the three points L , M and N ?

Chapter 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
 Coordinate geometry

Power plus

- 1 A line is drawn through the points $A(0, -2)$ and $B(3, 0)$. The x -coordinate of a point C on AB is 9. Find:
 - a the gradient of AB
 - b the equation of AB
 - c the y -coordinate of C .
- 2 The point $(-1, 6)$ lies on the line $kx + 3y - 13 = 0$, where k is a constant number. Find k .
- 3 $Z(-1, 3)$ is the midpoint of the interval joining $A(-4, 7)$ and B . Find the coordinates of B .
- 4 The circle has XY as a diameter and centre Z . What are the coordinates of X ?



- 5
 - a Find the gradient of any line parallel to $3x + 2y = 4$.
 - b Find the equation of the line that passes through the point $(0, -1)$ and is parallel to $3x + 2y = 4$.
- 6 $A(-1, 4)$, $B(4, 6)$, $C(2, 7)$ and D are the vertices of a parallelogram. Find the coordinates of D .

Chapter 3 review

Language of maths

axes	distance	exact answer	general form
gradient	gradient–intercept form	horizontal	interval
length	linear equation	midpoint	parallel
perpendicular	point–gradient form	reciprocal	rise
run	surd	vertical	x-axis
x-intercept	y-axis	y-intercept	

Puzzle sheet

Coordinate geometry
crossword

MAT10NAPS10014

- 1 What is the difference between the **y-axis** and the **y-intercept**?
- 2 When finding the length of an interval on a number plane, what is meant by an **exact answer**?
- 3 What measurement is the fraction given by the vertical rise of a line divided by the horizontal run?
- 4 What is the everyday meaning of the word **intercept**? Look it up in a dictionary.
- 5 What is the property of the gradients of perpendicular lines?
- 6 What form of a linear equation is $ax + by + c = 0$?

Topic overview

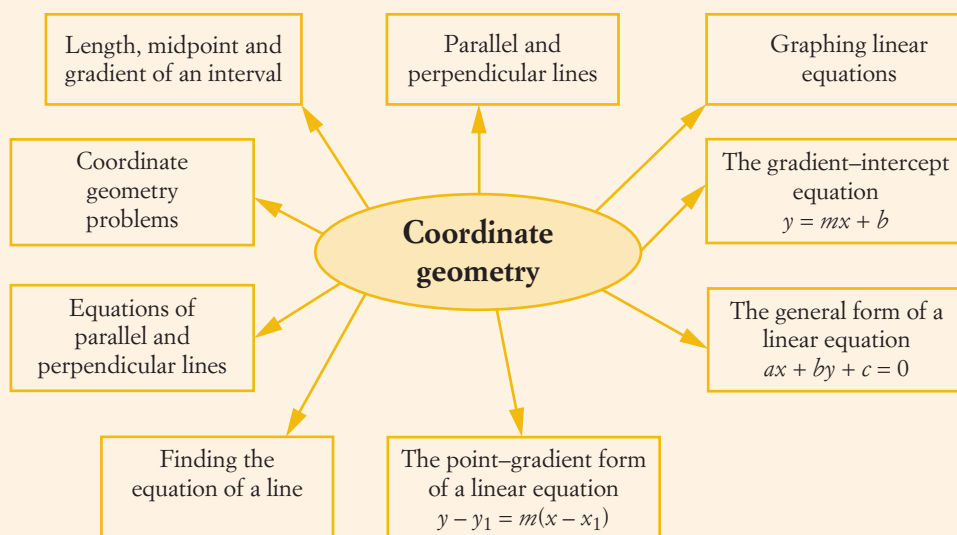
- How can you find the gradient of a line?
- When do you use the formula $y - y_1 = m(x - x_1)$?
- How can you test whether a pair of lines are perpendicular?
- What parts of this topic did you find difficult?

Quiz

Coordinate geometry

MAT10NAQZ00005

Copy and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Chapter 3 revision

See Exercise 3-01

- 1 An interval is formed by joining the points $K(5, 6)$ and $L(-7, 2)$.
 - a Find, correct to one decimal place, the length of interval KL .
 - b Find the midpoint of KL .
 - c Find the gradient of KL .

See Exercise 3-01

- 2 The vertices of a quadrilateral $HJKL$ are $H(-8, -5)$ $J(-1, -2)$ $K(2, 5)$ $L(-5, 2)$.
 - a Find the exact length of the sides of the quadrilateral.
 - b Find the gradient of each side of $HJKL$.
 - c Find the exact length of the diagonals HK and JL .
 - d What type of quadrilateral is $HJKL$?

Stage 5.3

See Exercise 3-01

- 3 Find, correct to the nearest degree, the angle of inclination of a line with gradient:
 - a 3
 - b $\frac{5}{4}$
 - c -1
 - d $-\frac{2}{3}$

See Exercise 3-02

- 4 A line passes through the points $V(8, -1)$ and $W(10, -2)$. What is the gradient of a line:
 - a parallel to VW ?
 - b perpendicular to VW ?

See Exercise 3-03

- 5 Graph each linear equation on a number plane.
 - a $y = -5x - 1$
 - b $x + 2y = 16$
 - c $3x + 4y - 12 = 0$

See Exercise 3-03

- 6 Test which of the following points lie on the line of $3x + y = 2$. Select the correct answer **A, B, C** or **D**.
 - A** (1, 0)
 - B** (2, 4)
 - C** (-1, 5)
 - D** (-1, -5)

See Exercise 3-03

- 7 What is the equation of the line through $(-2, 3)$ and parallel to the x -axis? Select the correct answer **A, B, C** or **D**.
 - A** $x = -2$
 - B** $x = 3$
 - C** $y = -2$
 - D** $y = 3$

See Exercise 3-04

- 8 Write the gradient, m , and y -intercept, b , for each linear equation.
 - a $y = 2x - 10$
 - b $y = 4x + 3$
 - c $y = \frac{4 - 3x}{8}$

See Exercise 3-05

- 9 Convert each equation to general form $ax + by + c = 0$.
 - a $y = 3x + 5$
 - b $y = \frac{2x}{5} - 10$
 - c $x = 3y + 6$

See Exercise 3-05

- 10 Rewrite each equation in the form $y = mx + b$, then state the value of the gradient, m , and the y -intercept, b .
 - a $x - y + 2 = 0$
 - b $2x - 8y + 8 = 0$
 - c $3x + y - 9 = 0$

Stage 5.3

See Exercise 3-06

- 11 Find, in general form, the equation of a line which passes through the point:
 - a (5, 5) and has a gradient of -3
 - b $(-1, 8)$ and has a gradient of $\frac{2}{5}$

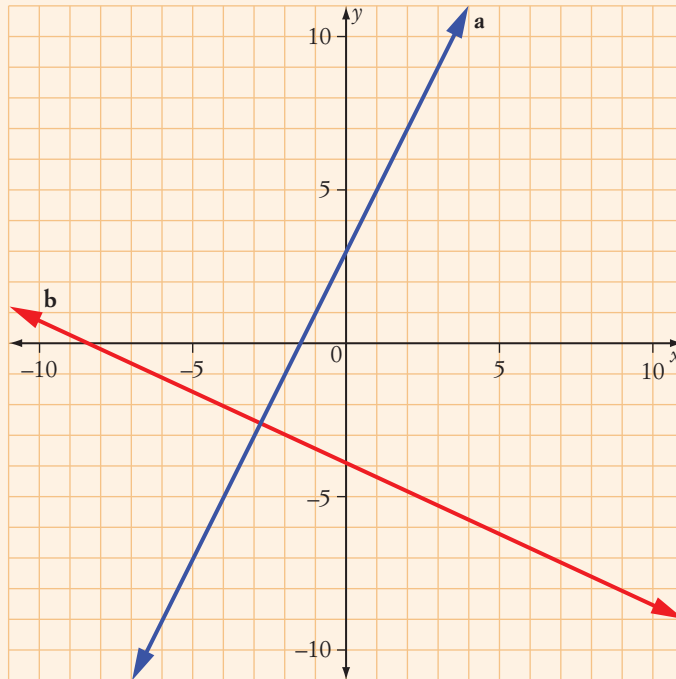
See Exercise 3-06

- 12 Find, in general form, the equation of a line which passes through the points:
 - a (10, 2) and (5, -1)
 - b $(-6, 3)$ and $(-2, -1)$

Chapter 3 revision

13 Find the equation of each line.

See Exercise 3-07



See Exercise 3-08

14 Find the equation of a line that is:

- a** parallel to $y = 3x + 1$ and passes through the x-axis at 2
- b** perpendicular to $y = \frac{x}{2}$ and passes through the origin.

15 The line $3x - 8y + 10 = 0$ and another line intersect at right angles at the point $(10, 5)$. Find the equation of the other line.

16 $L(-1, 2)$, $M(4, 5)$, $N(1, 10)$ and $P(-4, 7)$ are the vertices of a quadrilateral. Show that $LMNP$ is a square.

Stage 5.3

See Exercise 3-08

See Exercise 3-09